

Seaborn is an amazing data visualization library for statistical graphics plotting in Python. It provides beautiful default styles and colour palettes to make statistical plots more attractive. It is built on the top of the matplotlib library and also closely integrated to the data structures from pandas.

Install Seaborn

```
>pip install seaborn
```

Loading Built-in Datasets in Seaborn

Seaborn comes with several built-in datasets that are automatically downloaded and can be used for practice.

Listing All Available Datasets

```
import seaborn as sns
print(sns.get_dataset_names())
```

output:

```
['anagrams', 'anscombe', 'attention', 'brain_networks',  
'car_crashes', 'diamonds', 'dots', 'exercise', 'flights',  
'fmri', 'gammas', 'geyser', 'iris', 'mpg', 'penguins',  
'planets', 'taxis', 'tips', 'titanic']
```

Sample Code to Load and Display Dataset

Seaborn provides a dataset named **car_crashes**, which contains traffic accident statistics for US states. Let's load this dataset and view its structure.

Sample dataset: car_crashes

```
import seaborn as sns

# List all available datasets

print(sns.get_dataset_names())

# Load the 'car_crashes' dataset

df = sns.load_dataset('car_crashes')

# Display the first few rows

print(df.head())
```

Output:

```
total  speeding  alcohol  not_distracted  no_previous  ins_premium \
```

0	18.8	7.3	5.9	15.3	10.8	784.55
1	18.1	7.3	5.9	15.3	10.7	1053.48
2	18.6	6.9	5.9	15.9	10.6	899.47
3	22.4	8.5	5.9	18.8	11.3	827.34
4	12.0	4.2	3.4	10.3	7.9	878.41

```
ins_losses abbrev
```

0	145.08	AL
1	133.93	AK
2	139.91	AZ

3	142.39	AR
4	120.92	CA

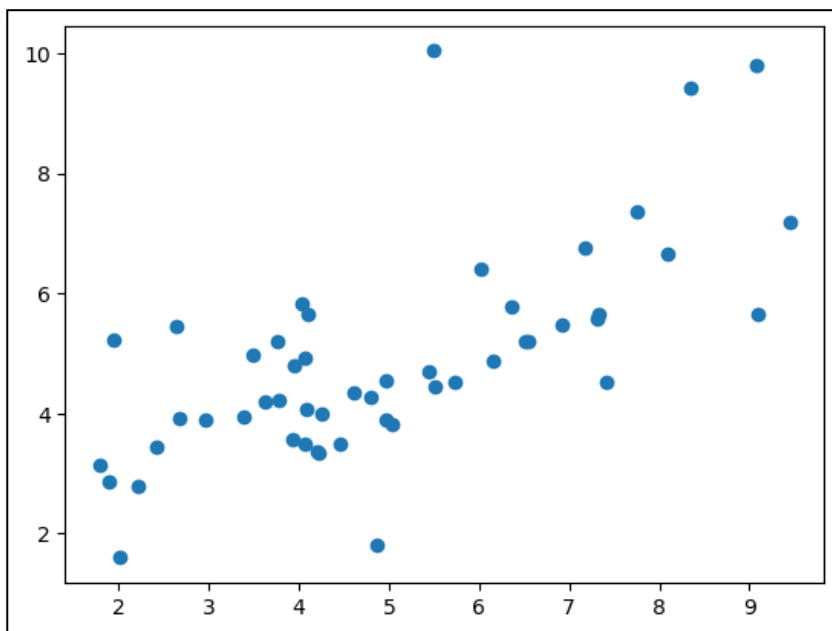
Styling and Themes We use visualizations to get insights from a huge amount of data or present our insights to the stakeholders and making these visualizations more pleasing can surely help with both of these. People are more likely to concentrate on beautiful and attractive visualizations rather than dull plots thus styling can be considered as a vital component of data visualization.

Matplotlib library is highly customizable, but it may be hard for us to tweak the right setting to get an attractive and good looking plot. Unlike Matplotlib, Seaborn comes packed with customized themes and a high-level interface for customizing and controlling the look of Matplotlib figures.

Customizing Plots with Seaborn

Basic Scatter Plot

```
import seaborn as sns
from matplotlib import pyplot as plt # alternate method
df = sns.load_dataset('car_crashes')
plt.scatter(df.speeding, df.alcohol)
plt.show()
```



Using `set_style()` to Apply Themes

Seaborn supports various themes that can make styling the plots really easy and save a lot of time. Using the `set_style()` function of Seaborn we can set any of the themes available on Seaborn library. Here are a few of the popular themes:

Darkgrid

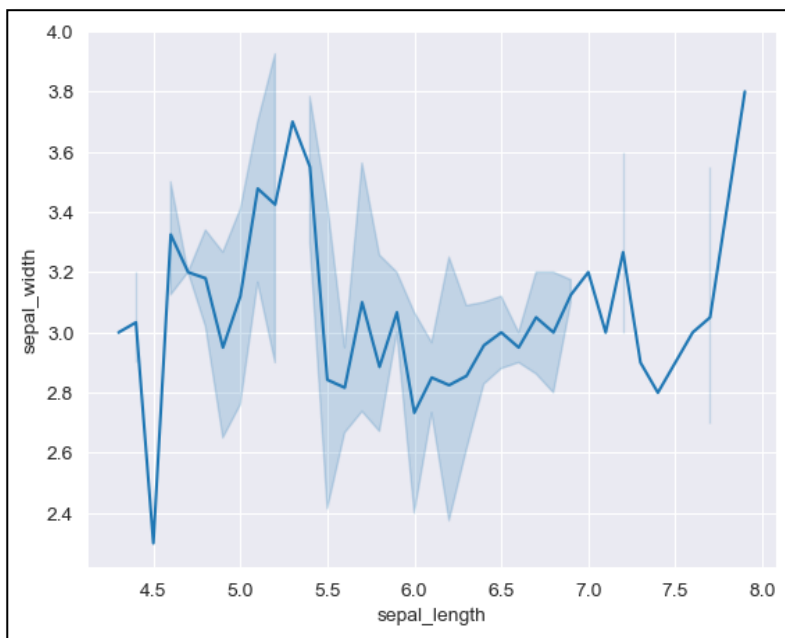
Whitegrid

Dark

White

Ticks

```
import seaborn as sns
import matplotlib.pyplot as plt
dataset = sns.load_dataset("iris")
sns.lineplot(x="sepal_length", y="sepal_width", data=dataset)
sns.set_style("darkgrid")
plt.show()
```



Combining Seaborn with Matplotlib

Adding Titles and Labels

Iris Dataset information

iris setosa



petal

sepal

iris versicolor



petal

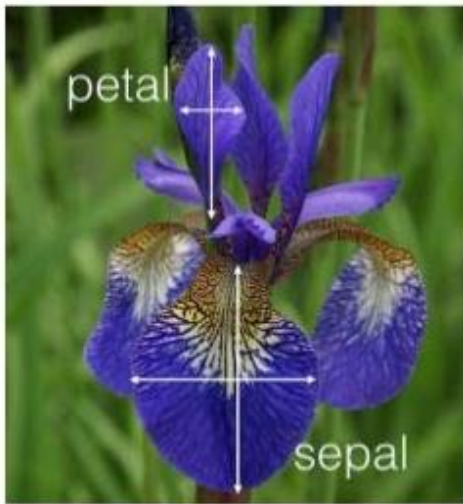
sepal

iris virginica



petal

sepal

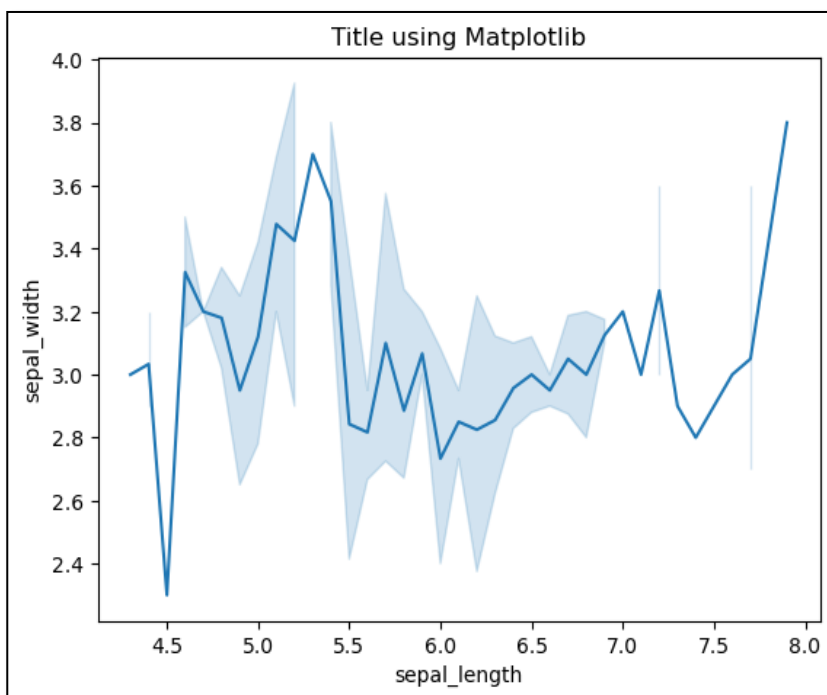


Training / test data				
Features				Labels
Sepal length	Sepal width	Petal length	Petal width	Species
5.1	3.5	1.4	0.2	Iris setosa
4.9	3.0	1.4	0.2	Iris setosa
7.0	3.2	4.7	1.4	Iris versicolor
6.4	3.2	4.5	1.5	Iris versicolor
6.3	3.3	6.0	2.5	Iris virginica
5.8	3.3	6.0	2.5	Iris virginica

Plot a line plot with sepal length in x axis and sepal width in y axis

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
data=sns.load_dataset("iris")
sns.lineplot(x="sepal_length",y="sepal_width",data=data)
plt.title('Title using Matplotlib')
plt.show()
```



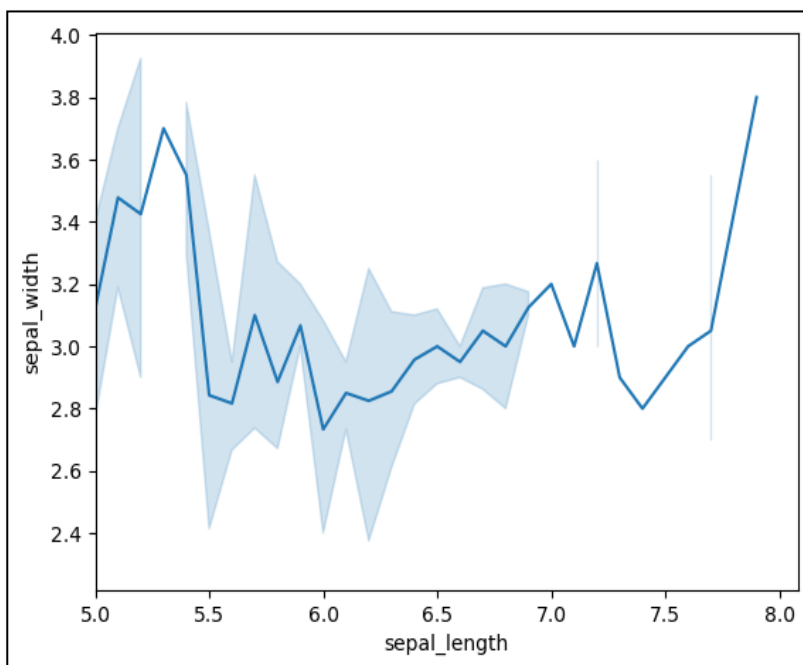
Setting Plot Ranges

```
# importing packages
import seaborn as sns
import matplotlib.pyplot as plt

# loading dataset
dataset = sns.load_dataset("iris")

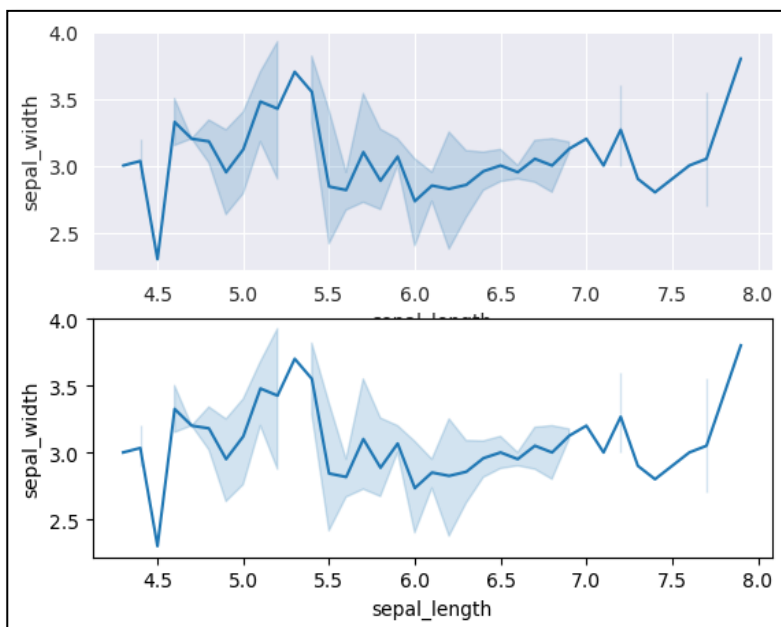
# draw lineplot
sns.lineplot(x="sepal_length", y="sepal_width", data=dataset)

# setting the x limit of the plot
plt.xlim(5)
plt.show()
```



Basic Subplots Example

```
# importing packages
import seaborn as sns
import matplotlib.pyplot as plt
# loading dataset
data= sns.load_dataset("iris")
with sns.axes_style('darkgrid'):
    plt.subplot(211)
    sns.lineplot(x="sepal_length",y="sepal_width",data=data)
plt.subplot(212)
sns.lineplot(x="sepal_length",y="sepal_width",data=data)
plt.show()
```



Tight Layout

Adjust the padding between and around subplots.

```
plt.tight_layout(*, pad=1.08, h_pad=None, w_pad=None, rect=None)
```

pad, default: 1.08

Padding between the figure edge and the edges of subplots, as a fraction of the font size.

h_pad, w_pad, default: pad

Padding (height/width) between edges of adjacent subplots, as a fraction of the font size.

rect (left, bottom, right, top), default: (0, 0, 1, 1)

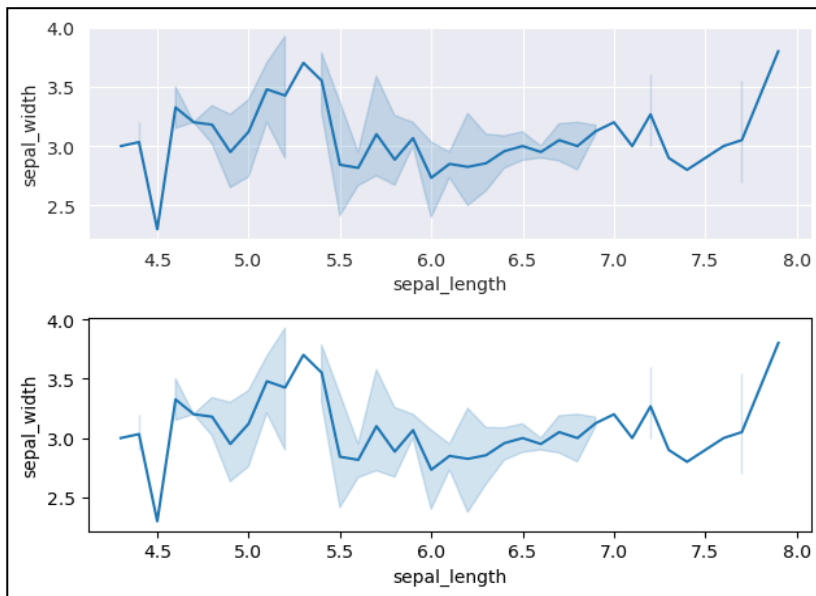
A rectangle in normalized figure coordinates into which the whole subplots area (including labels) will fit.

```
import seaborn as sns
import matplotlib.pyplot as plt

data= sns.load_dataset("iris")

with sns.axes_style('darkgrid'):
    plt.subplot(211)
    sns.lineplot(x="sepal_length",y="sepal_width",data=data)

plt.subplot(212)
plt.tight_layout()
sns.lineplot(x="sepal_length",y="sepal_width",data=data)
plt.show()
```



Color Palettes

Colormaps are used to visualize plots effectively and easily. One might use different sorts of colormaps for different kinds of plots. `color_palette()` method is used to give colors to the plot. Another function `palplot()` is used to deal with the color palettes and plots the color palette as a horizontal array.

Viewing Color Palette

```
import seaborn as sns
import matplotlib.pyplot as plt

# current color palette
palette = sns.color_palette()

# plots the color palette as a
# horizontal array

sns.palplot(palette)
```

```
plt.tight_layout()
plt.show()
```



Setting Default Palette

`set_palette()` method is used to set the default color palette for all the plots. The arguments for both `color_palette()` and `set_palette()` is same. `set_palette()` changes the default matplotlib parameters.

```
import seaborn as sns
import matplotlib.pyplot as plt
data = sns.load_dataset("iris")
# setting the default color palette
sns.set_palette('vlag')
plt.subplot(211)
# plotting with the color palette as vlag
sns.lineplot(x="sepal_length", y="sepal_width", data=data)
# setting another default color palette
sns.set_palette('Accent')
plt.subplot(212)
sns.lineplot(x="sepal_length", y="sepal_width", data=data)
plt.tight_layout()
plt.show()
```

